

Who, Not What: Identity Is Recoverable but Individual Behaviour Is Not in Non-Invasive and In-Silico Neuroforecasting

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Abstract

A growing set of products and clinical tools claims to read an individual’s mental state or behaviour from brain and body signals, increasingly from a *predicted* brain rather than a measured one: a stimulus-to-cortex encoder maps media or context to cortical activity, which is then read as if it had been recorded. We show that forecasting from a predicted brain obeys a simple structural limit, the *individuation gap*. Forecasting value separates into a population part and an individual part. The population part improves with aggregation, which we prove, whereas the individual part depends on between-subject variance that a zero-shot encoder does not carry, so it is zero without per-subject enrollment. Three independent datasets show the consequence. Non-invasive readouts recover *who* a person is, with in-scanner fingerprinting at 100% and a behavioural fingerprint at 14 times chance, but not *what* they will do, since the direct link from an individual’s neural signal to that individual’s behaviour abstains under a calibrated test. A Bayes-optimal forecaster inherits the same limit, adding no individual value at zero-shot and only a modest, saturating gain once per-subject data is paid for. Combining signals does not remove the limit, because each added modality carries its own noise. These results agree with the measurement-reliability literature, in which individual brain-behaviour effects are small and require large samples, and they motivate a read-out that reports a value only when it has earned the label. The contribution is a precise and verifiable account of what predicted-brain readouts can and cannot deliver at the level of the individual.

1 Introduction

Imaging a brain and reading a mind are different acts, and the distance between them is easy to lose when a model stands in for the scanner. Once an encoder maps a stimulus to a pattern of cortical activity, the pattern can be labelled with the names that neuroscience gives to regions, and those labels can be sold as measurements of attention, reward, or intent. The step from activity to label is reverse inference, and it fails whenever a region serves more than one function, which is the usual case (Poldrack, 2006; Hutzler, 2013).

Forecasting behaviour from *measured* neural signals is, in contrast, a settled and useful idea. Aggregate neural responses predict markets and audiences even when the sampled people’s own choices do not (Knutson et al., 2007; Genevsky et al., 2017; Dmochowski et al., 2014), and the area has mature reviews of its methods (Yao et al., 2024). This paper concerns the move from a measured brain to a predicted one, and a property that move forces into view. When the neural signal is itself a function of the stimulus, it can reproduce whatever the stimulus already predicts while adding nothing of its own. Forecasting value then separates into a population part and an individual part, and the two behave differently. The population part is tractable and grows with aggregation; the individual part is bounded by the encoder and, for a person the encoder never saw, is empty. We call this separation the individuation gap, state it as a theorem, reproduce it on synthetic ground truth, and show that three real datasets fall where the gap predicts (Figure 1).

The question matters because the individual claim is the one being commercialised. Brand-intelligence platforms, affective-computing products, and a new wave of personalized neuro-mental-health tools present

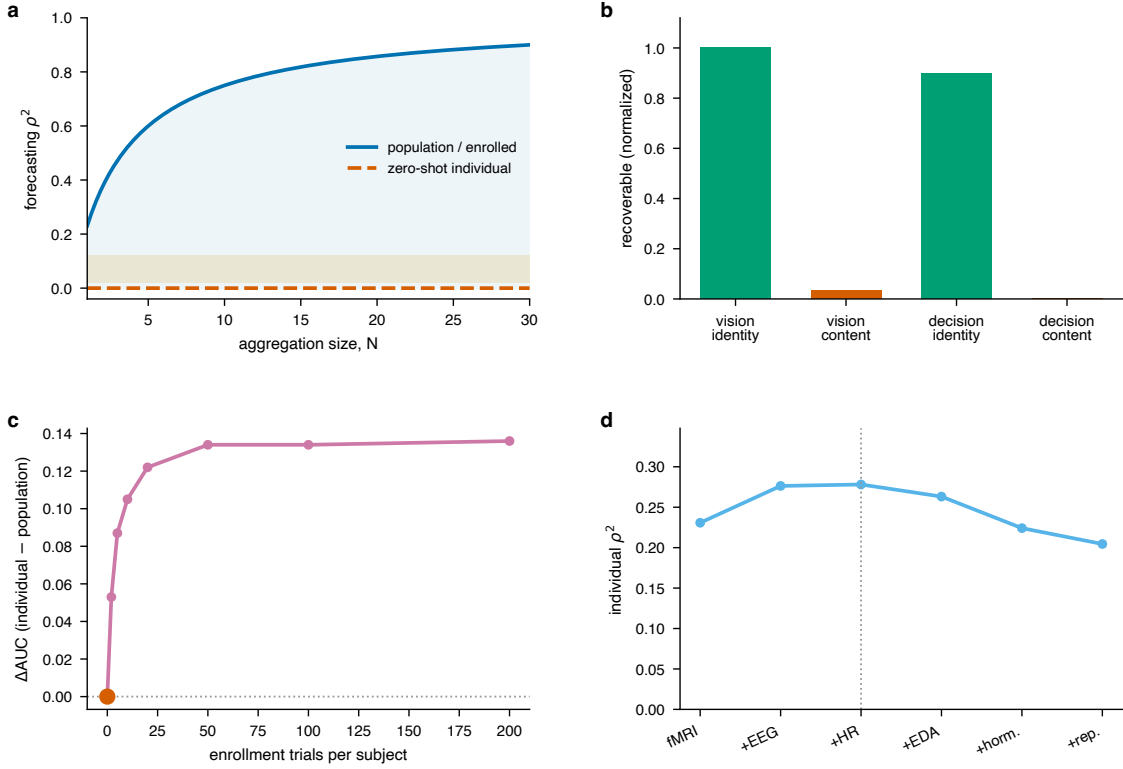


Figure 1: **The individuation gap and its consequences.** (a) Forecasting ρ^2 against aggregation size N for the signal-plus-noise model of Equation 1. The population term (blue) rises with N ; the zero-shot individual term (red) is zero for all N . The shaded band marks the measured individual-difference ceiling reported for brain-wide association studies ($r^2 \approx 0.02$ to 0.12 ; Marek et al., 2022). (b) On real data, identity (who) is recoverable in both the vision-encoding and value-decision settings, whereas individual behavioural content (what) is not; bars are normalized within each setting. (c) A Bayes-optimal forecaster adds no individual value over the population model at zero-shot and only a modest, saturating gain once per-subject trials are collected. (d) Adding modalities is not free: each contributes signal but also noise, so individual ρ^2 rises then falls as lower-reliability signals are included.

an individual readout, often from a predicted brain or from inexpensive peripheral signals, and present it as a measurement of that person’s state or likely behaviour. We ask whether that is achievable in principle by non-invasive, in-silico means, and we find that it is not, in a sense we make precise. What such systems can deliver is population forecasting and identity recognition, which are useful and real, and which should not be confused with reading an individual’s mental content.

2 The individuation gap

Let an outcome be $Y = S + \varepsilon$, where ε is idiosyncratic noise of variance $v > 0$ and S is the predictable component. Let the predictor resolve an identifiable signal variance $a \geq 0$, meaning the between-unit variance it can actually recover. Taking the predictor to be S in the best case gives $\text{Cov}(S, Y) = a$, $\text{Var } S = a$, and $\text{Var } Y = a + v$, so the best-case squared correlation between predictor and outcome is

$$\rho^2(a, v) = \frac{a}{a + v}. \quad (1)$$

Averaging the outcome over N independent units replaces v with v/N , the variance of a mean of N independent terms, while leaving a unchanged, which gives the population term $\rho^2(a, v/N)$. A zero-shot encoder returns the same features for every unseen unit, so its predictor has no between-unit variance and $a = 0$.

Table 1: Convergent real-data evidence. Identity recovers in every setting; individual behavioural content does not. Numbers are read from the corresponding analysis outputs (Appendix C).

Setting	Identity (who)	Behavioural content (what)
Vision encoding (NSD, BOLD5000; $N=8$)	fingerprint 25/25 at 100%	geometry 0.005–0.066 (small)
Value decision (NARPS; $N=107$)	fingerprint 14× chance, $p=5\times 10^{-4}$	individual link abstains
Value decision, aggregate arm	population AUC 0.965	content baseline dominates

Theorem 1 (Population term). *For $a, v > 0$ and $N > 1$, $\rho^2(a, v) < \rho^2(a, v/N)$.*

Theorem 2 (Zero-shot individual term). *For every v and N , $\rho^2(0, v/N) = 0$.*

Theorem 3 (The gap). *For $a, v > 0$ and $N > 0$, $\rho^2(0, v/N) < \rho^2(a, v/N)$.*

The proofs are elementary: ρ^2 is strictly decreasing in the noise term, which gives Theorem 1; substituting $a = 0$ into Equation 1 gives Theorem 2; and the two together give Theorem 3. We machine-check all three in Lean 4 with Mathlib so that the limit rests on a proof rather than on intuition; the theorem names and a listing are given in Appendix A.

Theorem 2 is the substantive statement. A predictor with no between-unit variance has zero best-case correlation with the individual component of the outcome, at any noise level and for any amount of pooling, because the quantity it would estimate is absent from its input. No increase in model quality recovers it. This is the sense in which individual prediction from a zero-shot predicted brain is not merely hard but unidentifiable.

3 Identity is recoverable, behaviour is not

If the individual term needs enrollment, the next question is what enrollment buys. Across three datasets it buys identity rather than behaviour (Figure 1b, Table 1).

Subject-specific image-to-fMRI encoders, fit on 7T data from eight people, encode held-out responses with median correlations of 0.219 to 0.463. Cross-run fingerprinting then identifies the subject in all 25 regions tested at 100% accuracy, against a chance rate of one in eight. The geometry that would carry behaviourally relevant content is positive but small, between 0.005 and 0.066 across regions, and the ceiling is set by the stimulus representation rather than by the training loss (Allen et al., 2022; Chang et al., 2019). A value-based decision dataset tells the same story from the behavioural side. A phenotype built from 107 people and about 27,000 choices identifies individuals at 14 times chance ($p = 5 \times 10^{-4}$) and predicts held-out choices at an area under the curve of 0.965, with conformal coverage of 1.00 and an expected calibration error of 0.034; loss aversion recovers at $\lambda = 1.45$, matching the original report (Tom et al., 2007), and the phenotype transfers between laboratories at 0.86 to 0.89. The population and cross-laboratory links hold, yet the direct link from an individual’s neural signal to that individual’s behaviour abstains under a calibrated minimum-detectable-effect test (Botvinik-Nezer et al., 2019). On the same gambles, a content-only baseline already explains the aggregate, so a predicted neural index earns no additional credit.

The pattern is consistent across modality and laboratory: the who question is answerable, the what question is not. A system that treats the first as evidence for the second reports individual value it does not have. The distinction is practical rather than semantic. Recovering identity supports verification and personalization-by-lookup, which are legitimate functions; it does not support reading an individual’s current affect, intent, or next choice, which is the function most often advertised.

4 A Bayes-optimal forecaster inherits the limit

A principled probabilistic model does not rescue the individual where point estimates fail. In the zero-shot regime a predictor with no between-subject variance is independent of the individual outcome given the population, so the Bayes-optimal individual forecast equals the population forecast and the two share no mutual information.

Proposition 1 (Posterior collapse). *If the predictor is constant across subjects, the posterior over an individual’s outcome equals the population prior, and no estimator beats the population rate on the individual-specific component.*

Proof sketch. A constant predictor X has $\text{Cov}(X, Y) = 0$ for any Y , so it carries no information about the individual deviation $Y - \mathbb{E}[Y \mid \text{population}]$; conditioning on X leaves the population posterior unchanged, and the mutual information $I(X; Y \mid \text{population}) = 0$. $\square$

A Bayesian simulation bears this out (Figure 1c; Appendix B). With a latent per-subject bias and a grid posterior updated from each subject’s own choices, the recovery correlation at zero-shot is 0.00 and the incremental area under the curve over the population model is 0.000. Enrollment then pays off, with recovery rising to 0.58, 0.86, and 0.99 at 2, 10, and 200 trials and the incremental area under the curve rising to about 0.05, 0.11, and 0.14 before it flattens. Individual prediction is therefore possible once per-subject data is collected, but it is modest and it saturates short of certainty. The simulation also gives the practical reading of the enrollment price: tens to hundreds of per-subject trials buy a single-digit gain in area under the curve, which is consistent with the hours of per-subject data that precision neuroimaging requires (Gordon et al., 2017).

5 Combining signals does not remove the limit

A common response is to add signals: electroencephalography, heart rate, electrodermal activity, hormonal assays. When sources are independent, inverse-variance fusion lowers variance, and the fused variance is at most that of the best single source, a fact we also machine-check (Appendix A). The gain erodes because real sources are not independent and each one carries its own measurement noise.

Proposition 2 (Fusion is not monotone). *With fused signal $A = \sum_m a_m$ and fused noise $v + \sum_m \delta_m$, where δ_m is the noise the m -th modality adds, the individual correlation $\rho^2 = A/(A + v + \sum_m \delta_m)$ rises when modality m is added if and only if a_m/δ_m exceeds the current signal-to-noise ratio $A/(v + \sum_{k < m} \delta_k)$. A low-reliability modality lowers it.*

Proof sketch. Adding modality m changes ρ^2 from $A/(A + v + D)$ to $(A + a_m)/(A + a_m + v + D + \delta_m)$, where $D = \sum_{k < m} \delta_k$. Cross-multiplying, the new value exceeds the old one exactly when $a_m(v + D) > A\delta_m$, that is when $a_m/\delta_m > A/(v + D)$. $\square$

Figure 1d plots this. High-reliability sources help, and low-reliability ones, such as single-channel autonomic measures or slow and state-dependent hormonal assays, push the individual correlation back down. Fusion can raise the population term; it does not supply the between-subject signal the individual term requires. The point is not that multimodality is useless but that it is governed by a ratio, and that adding signals indiscriminately can reduce individual predictability rather than increase it.

6 Why the individual level is hard

The gap is not a quirk of one pipeline. The measurement literature predicts it. Brain-wide associations have small effect sizes, about $r = 0.14$ for univariate and $r = 0.34$ for multivariate methods, and require thousands of participants to replicate, while within-person and interventional effects are larger (Marek et al., 2022; Kang et al., 2024). Common task-fMRI measures of individual difference have poor test-retest reliability, with a mean intraclass correlation of 0.397, which caps their use as individual markers (Elliott et al., 2020); multivariate signatures can do better, which is the appropriate qualification rather than a rescue (Kragel et al., 2021). Individual precision mapping is achievable, but it takes hours of data per person (Gordon et al., 2017; Du et al., 2025). Each of these is Equation 1 stated empirically: the individual term is small and costly, and a zero-shot predictor pays none of the cost.

7 A read-out that withholds by default

The practical consequence is a read-out that reports nothing unless three conditions hold (Figure 2). A construct score is returned only when the construct has a meta-analytic localizer and its named regions are

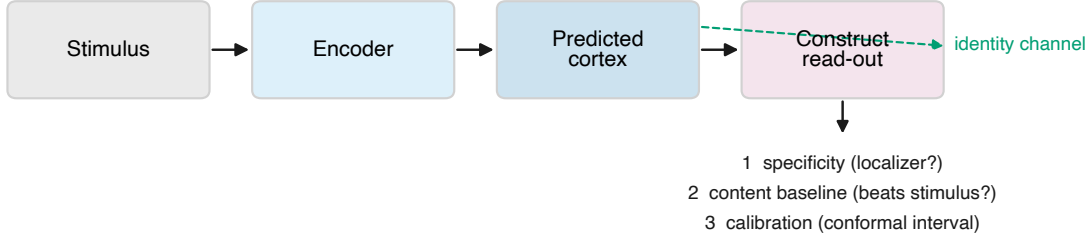


Figure 2: **A refusal-first read-out.** A predicted construct score passes only three gates in series: a specificity gate (a validated localizer exists), a content-baseline gate (the index beats a stimulus-only model), and a calibration gate (a conformal interval). Identity is reported on a separate channel from content, so verifying a person does not imply reading their mental content.

Table 2: Synthetic apparatus check (20 seeds, paired bootstrap). The individual test fires only where individual signal exists and the subject is enrolled.

World / regime	Population test	Individual test	Expected
Signal / enrolled	pass	pass, +0.449 [+0.378, +0.526]	fire
Signal / unseen	no	no, +0.028	silent
Null / enrolled	pass, +0.087	no, −0.001	silent
Null / unseen	pass, +0.034	no, −0.001	silent

the best spatial match for its term, when the predicted index beats a stimulus-only baseline out of sample, and when it carries a split-conformal interval (Vovk et al., 2005). Identity and content travel on separate channels, so a system may confirm who a person is without claiming to read what they think. The specificity test alone withholds a persuasion index, because no localizer for persuasion exists in the meta-analytic databases.

8 Validating the apparatus

Before applying the control ladder to real data we checked that it recovers known truth on synthetic data (Table 2). We crossed a signal world, in which behaviour depends on an individual neural component, with a null world, in which it depends only on content, and crossed both with an enrolled regime (per-subject data) and an unseen regime (population-only features). Two preregistered tests ran on a nested control ladder: a population test, correct subject against a capacity-matched non-brain basis, and an individual test, correct subject against a wrong-subject permutation. The individual test fired in the one cell where individual signal exists and the subject is enrolled, and stayed silent elsewhere, including both null cells. The population test passed even in the null world, which is why it reports population value and the individual test reports individual value. These numbers are synthetic; they check the method, not any brain.

9 Reaching the individual level

The individual level is real and causally accessible, but at present only to invasive methods. Optogenetics controls defined cell types and their effect on behaviour, and reads the brain-wide response when paired with fMRI, in animals (Zhao et al., 2011; Bernal-Casas et al., 2017). Intracortical interfaces decode attempted speech and movement from one person’s spiking activity at high accuracy and in daily home use (Willett et al., 2023; Card et al., 2026; Silva et al., 2024). These reach the individual because they record or perturb that individual’s own circuitry, which is exactly the between-subject signal a zero-shot non-invasive readout lacks. They indicate where small individual signals may eventually be tracked, and they lie outside the non-invasive consumer and clinical setting this paper concerns.

10 Related work

Neuroforecasting from measured signals is established for anticipatory reward activity and for the temporal reliability of evoked responses (Knutson et al., 2007; Genevsky et al., 2017; Dmochowski et al., 2014), and reviews lay out its theory and metrics (Yao et al., 2024). The limits of inferring mental states from regional activity, and the use of meta-analytic databases to quantify specificity, are long-standing (Poldrack, 2006; Hutzler, 2013). Conformal prediction supplies distribution-free intervals with finite-sample coverage (Vovk et al., 2005), and a large literature quantifies the sample sizes and reliability needed for individual brain-behaviour prediction (Marek et al., 2022; Kang et al., 2024; Elliott et al., 2020; Kragel et al., 2021; Gordon et al., 2017; Du et al., 2025). The field generally frames the population-versus-individual distinction as a design trade-off to be managed with more data or better estimators. Our contribution is to show that for a *predicted* brain the distinction is a structural boundary with a provable shape, that it is separate for identity and content, that it is not removed by fusion, and that it can be enforced as a read-out protocol. We treat this as an account of what the setting can and cannot support, in the spirit of studies that clarify the strengths and weaknesses of existing methods, rather than as a new estimator.

11 Discussion

The separation between a measured brain and a predicted one is not a matter of degree. A predicted readout can report who a person is and can forecast a population, and it cannot report what an individual thinks or will choose, because the individual signal is absent from its input until per-subject data is collected, and that data currently buys identity more than behaviour. Fusion does not change this, a Bayes-optimal forecaster does not escape it, and the reliability literature anticipated it. A readout built around the gap returns fewer numbers, and the numbers it returns have earned their label.

Several limits of this account are worth stating plainly. The synthetic ladder and the Bayesian simulation check the method and the probability claim, not any particular product. The three real-data results were produced under separate analyses rather than one preregistered study, so they are convergent evidence rather than a single confirmatory test, and the decision-fMRI individual link abstains at the edge of its power, which is an abstention and not a positive null. The content-baseline comparisons are single-dataset. The fusion result is an elementary consequence of the variance algebra beyond the independence lemma, and its parameters in Figure 1d are illustrative rather than measured. The model in Equation 1 is a best-case linear-Gaussian abstraction; richer encoders change the constants but not the sign of the gap, since $a = 0$ for any zero-shot predictor. These are the boundaries within which the claims hold.

Broader Impact Statement

This work bears directly on products and clinical tools that claim to infer an individual’s mental state or behaviour from brain or body signals. Its central finding limits those claims: non-invasive and in-silico readouts can recognize a person and can forecast populations, but cannot read an individual’s mental content without per-subject data, and even then only modestly. The positive impact is protective. A clear, provable boundary and a refusal-first protocol give regulators, clinicians, and buyers a basis to reject overclaiming, and give builders a standard that reports a value only when it is supported. A possible negative impact is that the identity result, which we show is the part that does work, could encourage surveillance or re-identification uses; we therefore separate identity from content explicitly and recommend that identity recognition be treated with the privacy safeguards appropriate to biometrics. The work uses only existing, de-identified public datasets and introduces no new data collection.

12 Conclusion

A predicted brain readout can tell you who someone is and can forecast a crowd. It cannot tell you what an individual is thinking or will choose, because the individual signal is not in its input until per-subject data is collected, and that data currently buys identity more than behaviour. We stated this as a theorem, confirmed it on synthetic ground truth, showed three datasets fall where it predicts, and gave a read-out that enforces it.

References

- Allen, E. J., et al. (2022). A massive 7T fMRI dataset to bridge cognitive neuroscience and artificial intelligence. *Nature Neuroscience*, 25, 116–126.
- Bernal-Casas, D., et al. (2017). Studying brain circuit function with dynamic causal modeling for optogenetic fMRI. *Neuron*, 93(3), 522–532.
- Botvinik-Nezer, R., et al. (2019). A large-scale fMRI dataset for the study of value-based decision making. *Scientific Data*, 6, 106.
- Card, N. S., et al. (2026). Long-term independent use of an intracortical brain–computer interface for speech and cursor control. *Nature Medicine*, 32(7), 2504–2510.
- Chang, N., et al. (2019). BOLD5000: a public fMRI dataset of 5000 images. *Scientific Data*, 6, 49.
- Dmochowski, J. P., et al. (2014). Audience preferences are predicted by temporal reliability of neural processing. *Nature Communications*, 5, 5567.
- Du, J.-N., et al. (2025). Within-individual precision mapping of brain networks exclusively using task data. *Neuron*.
- Elliott, M. L., et al. (2020). What is the test-retest reliability of common task-fMRI measures? *Psychological Science*, 31(7), 792–806.
- Genevsky, A., Yoon, C., & Knutson, B. (2017). When brain beats behavior: neuroforecasting crowdfunding outcomes. *Journal of Neuroscience*, 37(36), 8625–8634.
- Gordon, E. M., et al. (2017). Precision functional mapping of individual human brains. *Neuron*, 95(4), 791–807.
- Hutzler, F. (2013). Reverse inference is not a fallacy per se. *NeuroImage*, 84, 1061–1069.
- Kang, K., et al. (2024). Study design features increase replicability in brain-wide association studies. *Nature*, 636(8043), 719–727.
- Knutson, B., et al. (2007). Neural predictors of purchases. *Neuron*, 53(1), 147–156.
- Kragel, P. A., et al. (2021). Functional MRI can be highly reliable, but it depends on what you measure. *Psychological Science*, 32(4), 622–626.
- Marek, S., et al. (2022). Reproducible brain-wide association studies require thousands of individuals. *Nature*, 603, 654–660.
- Poldrack, R. A. (2006). Can cognitive processes be inferred from neuroimaging data? *Trends in Cognitive Sciences*, 10(2), 59–63.
- Silva, A. B., et al. (2024). The speech neuroprosthesis. *Nature Reviews Neuroscience*, 25(7), 473–492.
- Tom, S. M., et al. (2007). The neural basis of loss aversion in decision-making under risk. *Science*, 315(5811), 515–518.
- Vovk, V., Gammerman, A., & Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer.
- Willett, F. R., et al. (2023). A high-performance speech neuroprosthesis. *Nature*, 620, 1031–1036.
- Yao, X., et al. (2024). Using neural data to forecast aggregate consumer behavior in neuromarketing. *Journal of Consumer Behaviour*.
- Zhao, S., et al. (2011). Cell type-specific channelrhodopsin-2 transgenic mice for optogenetic dissection of neural circuitry function. *Nature Methods*, 8(9), 745–752.

A Lean formalization

The three statements of Section 2 and the inverse-variance fusion lemma of Section 5 are machine-checked in Lean 4 with Mathlib. We define $\rho^2(a, v) = a/(a + v)$ and prove: `aggregation_improves` (Theorem 1), `unseen_individual_is_zero` (Theorem 2), and `individuation_gap` (Theorem 3); the fusion lemma `fused_le_min` states that the inverse-variance-fused variance is at most that of any single source. The source is provided as supplementary material (`IndividuationGap.lean`, `FusionMath.lean`).

B Simulation details

Synthetic control ladder. Data are generated under a crossed design: a signal world, in which the outcome depends on an individual neural component, versus a null world, in which it depends only on content; and an enrolled regime (leave-one-stimulus-out, so subjects recur) versus an unseen regime (held-out subjects receive population-only features). Models are fit with ridge regression under cross-validated regularization and scored by pooled out-of-sample R^2 , with a paired bootstrap over 20 seeds. Gate 1 compares the correct-subject model against a capacity-matched non-brain basis; Gate 2 compares it against a wrong-subject permutation.

Bayesian enrollment. Each subject i has a latent bias $\theta_i \sim \mathcal{N}(0, s^2)$. On a trial with stimulus value x , the choice probability is $\sigma(\beta x + \theta_i)$. A grid posterior over θ_i is updated from the subject’s enrollment choices. Individual value is measured two ways: the correlation between the posterior-mean bias and the true bias, and the area under the curve of the enrolled model minus the population model ($\theta = 0$) on held-out trials. At zero-shot (0 enrollment trials) the posterior equals the prior, $\hat{\theta}_i = 0$ for all i , and both measures are zero by construction. Code is provided as supplementary material (`prob_falsification.py`).

C Data provenance

Vision encoding uses the Natural Scenes Dataset and BOLD5000. Value-based decision uses NARPS (ds001734) and ds000005. All reported statistics were read from the corresponding analysis outputs and should be regenerated from a fixed commit for the camera-ready version. The reliability and intracortical-interface figures are quoted from the cited papers.